



Form3
Design
Design and Development of a WhatsApp-
Based Issue Reporting and Analytics
Dashboard System for TRAMOS Web at PT
Andakara Informatika Nusantara

GROUP MEMBER:

No.	Student Name	Student ID	Study Program
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Submitted for
Capstone Design Project
to Faculty of Computer Science
President University

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STATEMENT OF ORIGINALITY

In my capacity as an active student at President University and as the author of the Capstone Design Project stated below:

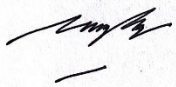
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Faculty : Computer Science

I hereby declare that my Capstone Design Project entitled “**Design and Development of a WhatsApp-Based Issue Reporting and Analytics Dashboard System for TRAMOS Web at PT Andakara Informatika Nusantara**” is to the best of my knowledge and belief, an original piece of work based on sound academic principles. If there is any plagiarism detected in this final project, I am willing to be personally responsible for the consequences of these acts of plagiarism and will accept the sanctions against these acts in accordance with the rules and policies of President University.

I also declare that this work, either in whole or in part, has not been submitted to another university to obtain a degree.

Cikarang, 1 April 2026

Signer 1	Signer 2
	
Cayla Winri Azzahra - 012202300098	Ivander Johana Pratama - 012202300030

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Interface Between Syst...

Hybrid Agile Methodolo...

Detail of Components a...

UML Diagram (Class / S...

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Form3

Capstone Design Proposal

Design and Development of a

WhatsApp-Based Issue Reporting and

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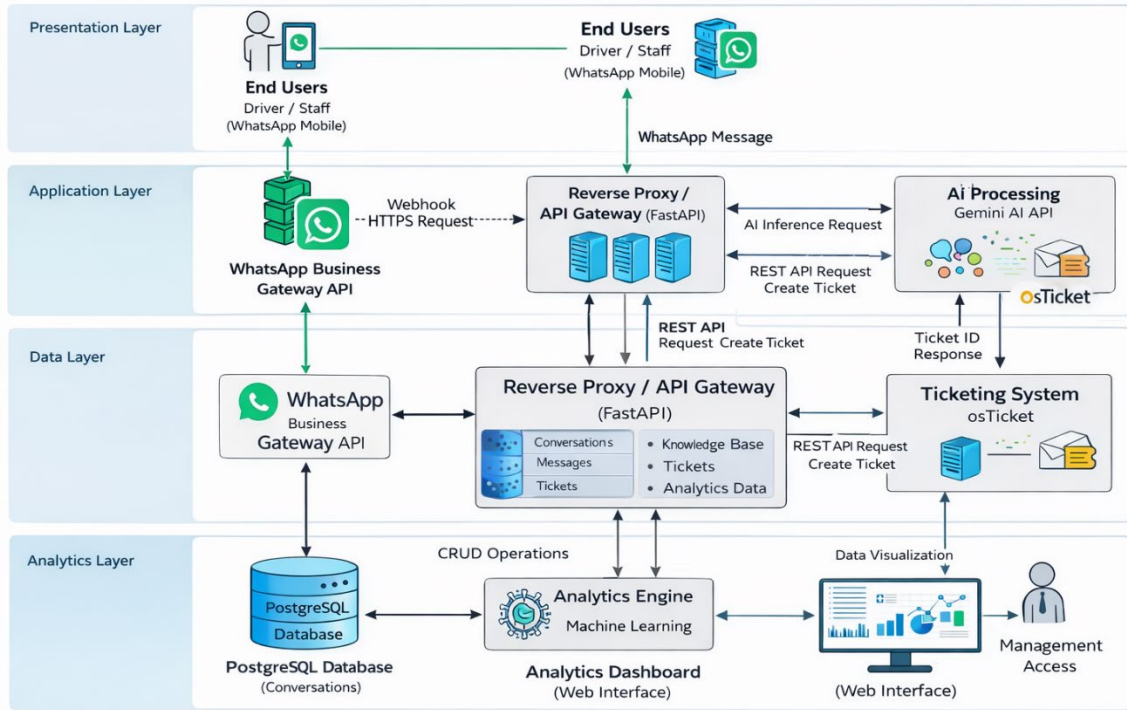
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A. SYSTEM DESIGN



The system developed in this project has been implemented by considering a service-oriented architecture (SOA) design, where the backend services are modular in nature, incorporating various important components like the WhatsApp communication interface, backend API processing, artificial intelligence (AI) analysis, ticket management, and analytics. The overall design enables the system to perform the issue reporting functionality in an automated manner.

On the user side, the users, either being a driver or operational personnel, would interact with the system by making use of the WhatsApp application on their mobile device to report issues encountered in the TRAMOS system. The messages sent by users to the WhatsApp application would be received by the WhatsApp Business Gateway API, which would forward the messages to the backend of the system by making use of the HTTPS Webhook protocol. The backend of the system has been implemented by making use of FastAPI, which acts as an API Gateway and Reverse Proxy.

Once the message is received by the backend, it will be forwarded to the AI Processing module, which will use the Gemini AI API to analyze the message sent by the user. This includes analyzing the intent, categorization of the issue, and checking the knowledge base to find a solution. If a solution exists, it will be used to respond automatically to the user's message on WhatsApp. However, in case the issue cannot be resolved, a REST API request will be sent to the osTicket ticketing system to automatically create a support ticket.

All the conversations, messages, tickets, and issue resolution results will be saved in a PostgreSQL database, which will act as a repository of information. The information saved in

the database will be processed using an analytics engine, which will perform a series of statistical calculations using machine learning-based analytics. The results obtained from the analytics will be displayed in a web-based analytics dashboard, which can be accessed by the management.

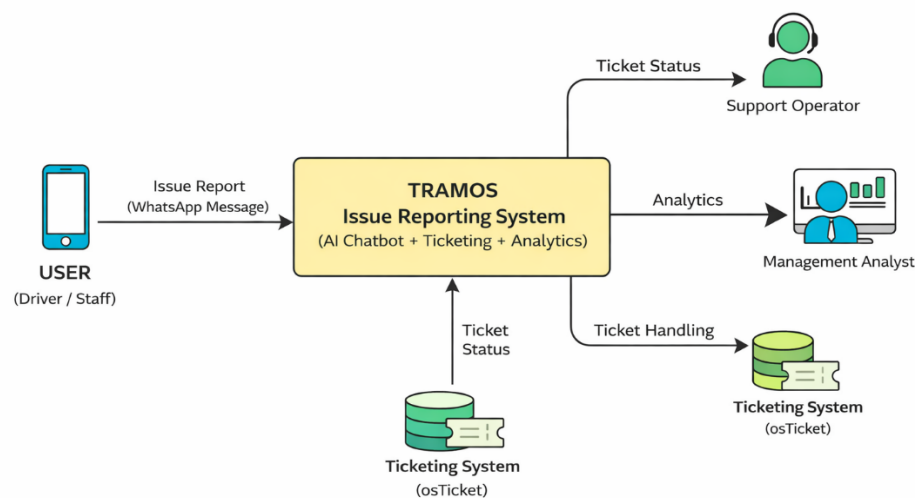
Thus, the system has the potential to provide a real-time, automated, and well-documented issue reporting system, which can be scaled up according to the needs of the TRAMOS system.

B. HIERARCHICAL/ITERATIVE DESIGN

1. Data Flow Diagram of TRAMOS Issue Reporting System

This section shows the hierarchical structure of the system using Data Flow Diagrams (DFDs) from level 0 to level 2. Data Flow Diagrams show the flow of data within the TRAMOS Issue Reporting System, starting with the general interaction of the system (level 0), then moving to the general processes within the system (level 1), and lastly to the internal processes within the system (level 2). This hierarchical structure of the system shows a better understanding of the system.

A. Data Flow Diagram Level 0 (Context Diagram)

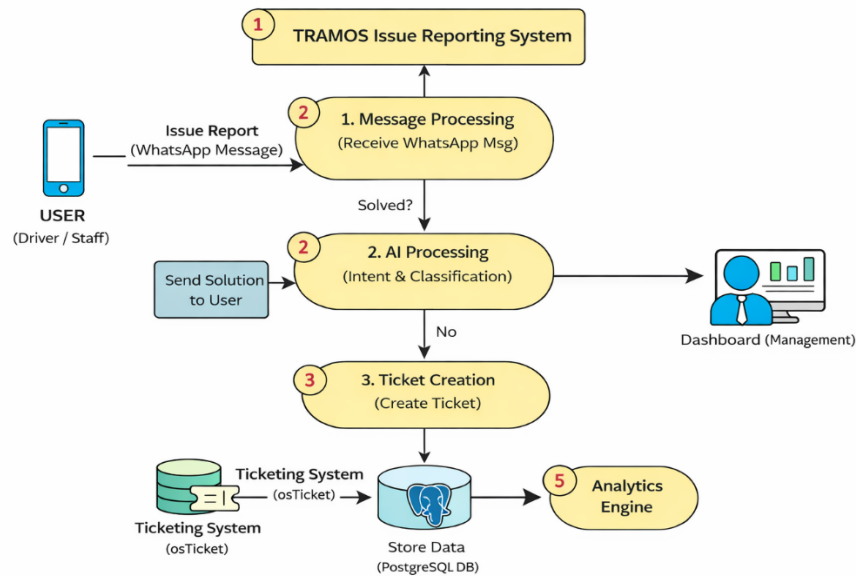


Data Flow Diagram (DFD) Level 0 depicts the overall interaction between the TRAMOS Issue Reporting System and the external entities. In this context, the entire system can be considered a process, which includes issue reporting, ticket management, and analytics processing.

The external entities are the user, who can be either a driver or a staff member, sending issue reports in the form of WhatsApp messages to the system for processing. The system also processes the reports, which can result in ticket information, sent to the ticket management system (osTicket) for further processing.

The ticket management system then sends ticket information to the support operator, which can be used for monitoring the issues, as well as sending analytics information to the management analyst for decision-making purposes.

B. Data Flow Diagram Level 1

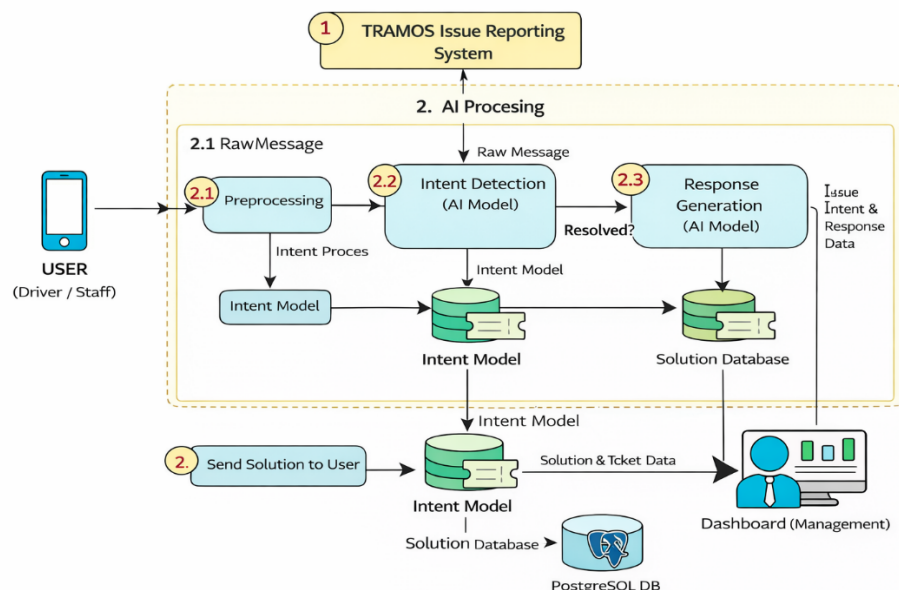


Data Flow Diagram (DFD) Level 1 for the TRAMOS Issue Reporting System shows how the internal processes of the system are decomposed. The process starts when the user sends an issue report through WhatsApp, which is received and forwarded by the Message Processing component.

Next, the message is analyzed through the AI Processing stage, where the intent of the user is detected and classified. If the issue is solvable, a solution is sent back to the user. However, if the issue is not solvable, the process proceeds to the Ticket Creation module, where a ticket is created and sent to the ticketing system (osTicket).

All the data regarding messages, tickets, and user interactions is stored in the PostgreSQL database, which is processed by the Analytics Engine for generating insights. These insights are shown through a dashboard that can be accessed by the management for easier monitoring.

C. Data Flow Diagram Level 2



The Data Flow Diagram (DFD) at Level 2 shows the internal process of the AI Processing module of the TRAMOS Issue Reporting System. The process starts when a user sends a message. The message is first preprocessed to clean and normalize the raw input data.

Next, the preprocessed data is analyzed by the intent detection module of the AI model. The intent of the user and the type of issue are identified. Based on this information, the user is directed to the response generation module. In this module, a suitable solution to the issue is retrieved from the solution database.

If the issue has been resolved, the solution is sent back to the user. Otherwise, the preprocessed data, including user intent and issue details, is stored in the database. The necessary information is also forwarded to the management dashboard.

2. Interface Between System Components

The TRAMOS Issue Reporting System employs a mix of message passing, parameter passing, and shared data storage as a means of communication between system components. Message passing is mainly employed for communication between external services such as WhatsApp Gateway, AI processing modules, and the ticketing system. On the other hand, parameter passing is mainly employed for communication between internal processes for the exchange of structured information such as user messages and tickets. Shared data storage is also employed through the PostgreSQL database for the purposes of consistency and synchronization.

1. Message Processing Interface

This module is responsible for the receipt of incoming messages from the user through the WhatsApp platform.

a. Receive Message (Message Passing)

This involves the receipt of a message from the user through the WhatsApp Gateway. The message is received through a webhook request. The message details are then sent to the backend as a structured message.

b. Forward Message (Parameter Passing)

This involves the forward of a message to the AI Processing module as a parameter.

2. AI Processing Interface

This module is responsible for processing and analyzing messages from users using AI.

a. Preprocessing (Parameter Passing)

The message is preprocessed and cleaned before it is passed to the AI model for analysis.

b. Intent Detection (Message Passing)

The preprocessed message is passed to the AI model (Gemini API) to detect user intent and categorize the issue category. The result is obtained as a response message.

c. Response Generation (Message Passing)

If a solution is obtained, relevant information is retrieved from the solution database and a response message is generated and sent back to the user.

3. Ticket Creation Interface

This module deals with escalation in cases where the problem cannot be resolved programmatically.

a. Collect Issue Data (Parameter Passing)

The user input for problem description, vehicle unit, and location is passed as parameters.

b. Create Ticket (Message Passing)

A REST API request is sent to the osTicket system with structured data for the ticket (subject, description, category, and priority).

c. Receive Ticket Response (Message Passing)

The osTicket system sends back a ticket ID to be passed to the user and stored in the database.

4. Data Storage Interface

This module facilitates data persistence and sharing.

a. Store Data (Parameter Passing)

System data, which includes messages, conversations, tickets, and resolution, are passed as parameters and stored in the PostgreSQL database.

b. Data Retrieval (Global Data Access)

Other modules can retrieve the stored data from the database as and when needed, for example, conversation history for AI or ticket information for analytics.

5. Analytics Interface

This module is used for data processing for insights and reporting.

a. Data Processing (Parameter Passing)

Data is passed from the database for aggregation and analysis by the analytics engine.

b. Dashboard Communication (Message Passing)

Data is passed from the analytics engine for display on the dashboard interface for management users.

3. Hybrid Agile Methodology for System Development

In developing the TRAMOS Issue Reporting System, the Hybrid Agile approach will be used. This approach will integrate elements of the Scrum methodology, Prototyping methodology, and Incremental Development methodology. This approach will be used in developing the system due to the presence of AI elements and integrations with other systems like WhatsApp API and osTicket.

This approach, unlike the pure Scrum methodology, allows for flexibility in experimenting with AI models, yet it provides structured development cycles. It also allows for continuous improvement, iterative testing, and gradual system integration.

- Project Management Approach

The development process is an iterative process in which the system is developed and enhanced step by step. Rather than adhering to a strict role-based approach as defined in Scrum development methodologies, the development team works in a fluid manner focusing on achieving the tasks and making the system functional. The development process includes requirement analysis, system development, testing, and refinement of the system. Regular evaluation of the development progress is done to match the system development and design.

- Database Design

The database is designed incrementally based on the system's requirements and the data flow within the system. To begin with, entities such as users, conversation history, and tickets are defined in the database. As the system is enhanced and new features are added to the system, entities such as analytics data, message history, and artificial intelligence-related data are added to the database.

- AI & System Development

The development of the AI component takes place using the prototyping method, where various models and logic are implemented. The system processes the messages sent by the users through pre-processing, intention detection, and response generation. The chatbot's performance improves over time based on the results obtained during the testing phase.

- System Integration

The system integrates several components, including the WhatsApp gateway, AI, database, and osTicket. The system integrates the components incrementally, starting from the simplest component, i.e., the message component, followed by the AI component, then the ticket component, and finally the analytics component. The components' integration takes place through the testing phase to ensure proper communication between the components and reduce system error.

- Testing and Refinement

The testing phase takes place continuously during the development phase. Every module is individually tested, and then the modules are integrated into the system. The feedback obtained from the testing phase helps refine the system's functionality, improve its performance, and correct the errors.

4. Detail of Components and Libraries Used

To develop and implement the TRAMOS Issue Reporting System, various hardware, software, AI services, databases, and development tools are employed. These components play a crucial role in ensuring the system is efficiently functional, enables real-time communication, and is able to process information in a scalable manner.

1. Hardware

- a. Laptop / Mobile Device

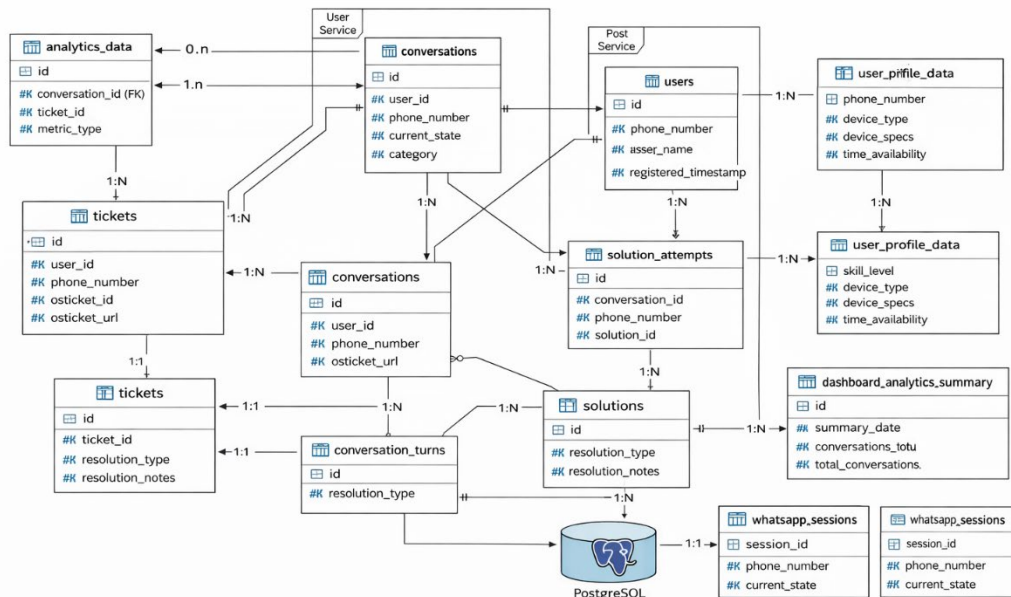
The system can be accessed through laptops or mobile devices. The users, i.e., the drivers and staff, communicate with the system through WhatsApp on their mobile phones. On the other hand, the operators and management communicate through the system by accessing the web-based dashboards through laptops or desktop computers.

2. Software

- a. Python (FastAPI)
FastAPI is the primary backend framework for developing RESTful APIs for processing messages and efficiently communicating with AI services and the system.
- b. WhatsApp Gateway (Webhook API)
The Webhook API is employed for receiving and sending messages to the users and the system. This is the primary communication channel for users to report issues to the system.
- c. REST API
The system employs RESTful APIs to integrate various components of the system, such as the chatbot, database, analytics tool, and ticketing system.
- 3. AI and External Services
 - a. AI Language Model (Gemini API)
Used for processing user messages, detecting user intent, and generating responses based on the reported problem.
 - b. osTicket System
An external ticketing system for handling tickets. The integration of the ticketing system is done using API calls to the osTicket system for creating tickets.
 - c. Ngrok (Development Tool)
Used for allowing the local backend server access to the internet for integrating the webhook during development and testing.
- 4. Database and Data Handling
 - a. PostgreSQL
Used for storing structured data such as user information, conversation history, tickets, and analytics information.
 - b. JSON Data Handling
Used for handling unstructured data such as AI-generated responses and conversation history within the application.
- 5. Development and Utilities
 - a. Visual Studio Code
This is used as a code editor for the development of the backend logic of the system integration.
 - b. Postman
This tool is used for testing the API endpoints of the system.
 - c. GitHub
This tool is used for version control of the code changes by team members during the development of the system.
 - d. Figma
This tool is used for designing the interface of the system.

5. UML Diagram (Class / Sequence / Activity)

1. Entity Relationship Diagram

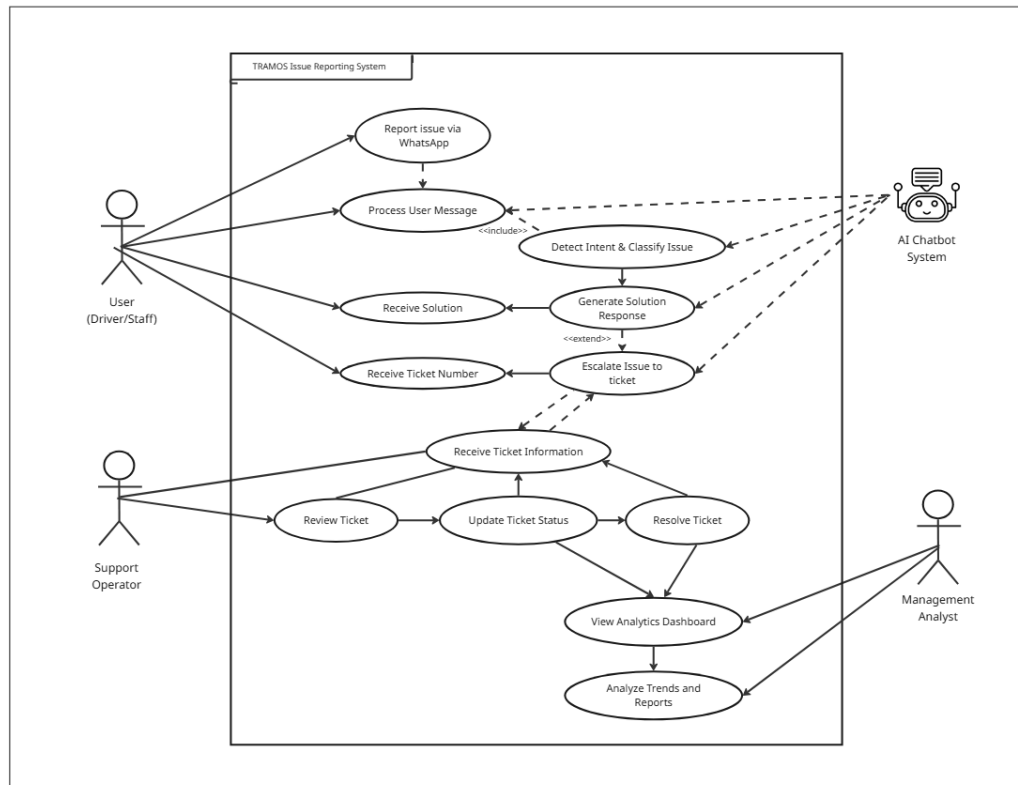


- Users (User Entity)**
 This table contains all user-related data, including phone number, name, and language. Each user is uniquely identified by their user ID. Users are essentially the main actors that interact with our system by sending issue reports in the form of WhatsApp messages. A user may have multiple conversations and create multiple support tickets.
- Conversations (Conversation Entity)**
 This table represents all communication sessions that occur between users and our system. Each conversation is associated with a user. Each conversation contains data such as phone number, state, and issue category. Conversations act as the primary entity that contains messages exchanged, AI processing, and ticket creation.
- Message History (Message Entity)**
 This table contains all messages that have been exchanged in any conversation. Each message is associated with a conversation. Each message is essentially used for tracking purposes.
- Conversation Turns (Interaction Steps Entity)**
 This table tracks the step-by-step interaction within a given conversation. A given conversation may have many steps, e.g., greeting, describing the issue, attempting to solve it, and finally confirming that it is resolved.
- Tickets (Ticketing Entity)**
 This table tracks tickets that are created in case issues cannot be automatically resolved by the AI. A given ticket is associated with only one user and one conversation. The information tracked includes the ID, phone number, and osTicket reference.
- Resolutions (Resolution Entity)**

This table tracks information about how each given ticket is resolved. A given ticket is associated with only one resolution, which may include information like type and notes. The resolution may be done by the AI or human operators.

7. **Solution Attempts (AI Solution Entity)**
This table stores information about all the solution attempts generated by the system during a conversation. A conversation may have many solution attempts, as it involves various troubleshooting steps provided by the AI.
8. **Analytics Data (Analytics Entity)**
This table stores analytical data generated from conversations and tickets. This data may include information like issue category, resolution time, and system performance, etc.
9. **Conversation Context (Context Entity)**
This entity stores information about the state of the current conversation, like issue description and context. This is used for real-time interaction and for ongoing conversations.
10. **WhatsApp Sessions (Session Entity)**
This table is used to store active user sessions in WhatsApp interactions. Each user may have multiple sessions, which represent different activities in issue reporting.
11. **User Profile Data (User Profile Entity)**
This table is used to store other user data, including skill levels, device types, and interaction types. This table is used in personalizing system responses and enhancing AI model performance.
12. **Solution Effectiveness (AI Performance Entity)**
This table is used to store the effectiveness of solutions provided by the AI system. The table is used in tracking how effective each solution is, based on previous attempts. This is used in enhancing the AI model.
13. **Dashboard Analytics Summary (Summary Entity)**
This table is used in storing summarized analytics data. This includes daily reports on user conversations, tickets, and system performance. The table is used in supporting management in making appropriate decisions.

2. Use Case Diagram



This use case diagram illustrates how users, support operators, management analysts, and the AI chatbot interact with the TRAMOS Issue Reporting System. The system is divided into three main functional categories:

1. Issue Reporting & AI Processing

This section describes the process of issue reporting and how the AI system processes the issues:

- **Report Issue via WhatsApp:** This is where the user (driver/staff) can report the issues using the WhatsApp messages.
- **Process User Message:** The system processes the message sent by the user.
- **Detect Intent & Classify Issue:** The AI chatbot will analyze the message sent by the user to determine the type of issue being reported.
- **Generate Solution Response:** Depending on the intent of the message, the AI generates the appropriate solution to the issue.
- **Receive Solution:** The user receives the solution to the issue directly from the system.
- **Escalate Issue to Ticket:** If the issue is not easily solved by the AI, the system will escalate the issue to the support ticket feature.

2. Ticket Management System

This section describes the process of how the support ticket is handled by the system and the operators:

- **Receive Ticket Number:** The user receives the ticket number to be used for tracking purposes.
- **Receive Ticket Information:** The user and operator can receive the ticket information.
- **Review Ticket:** The operator reviews the details of the reported issue.

- Update Ticket Status: The operator can update the status of the support ticket.

Resolve Ticket: The operator provides the resolution to the reported issue.

3. Analytics & Monitoring

This part of the presentation is about monitoring the system and making decisions:

- View Analytics Dashboard: Management analysts can view the performance of the system through a dashboard.
- Analyze Trends and Reports: Management analysts can analyze the trends of issues, resolutions, and performance of the system.

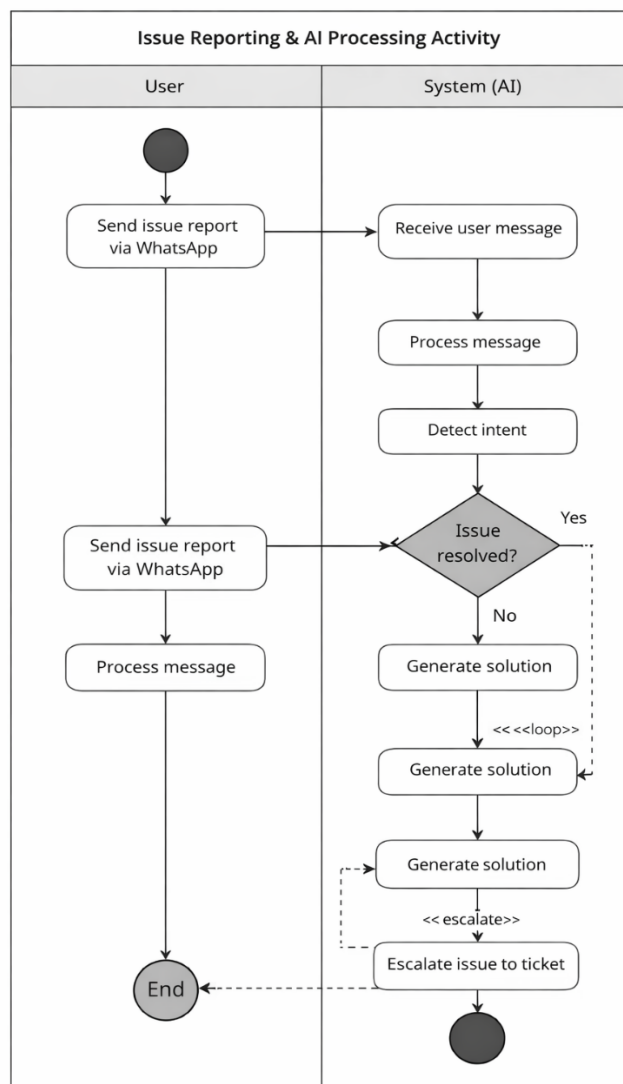
4. Role of AI Chatbot

The role of the AI Chatbot System is that of a "supporting actor" in:

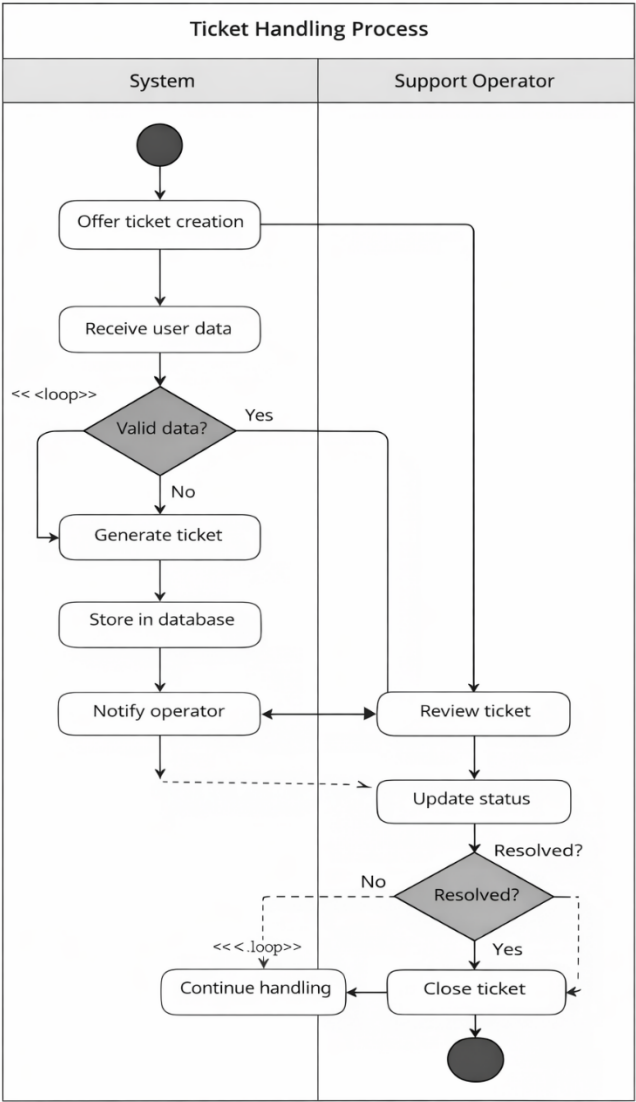
- Processing user messages
- Identifying intent
- Sending automated responses

3. Activity Diagram

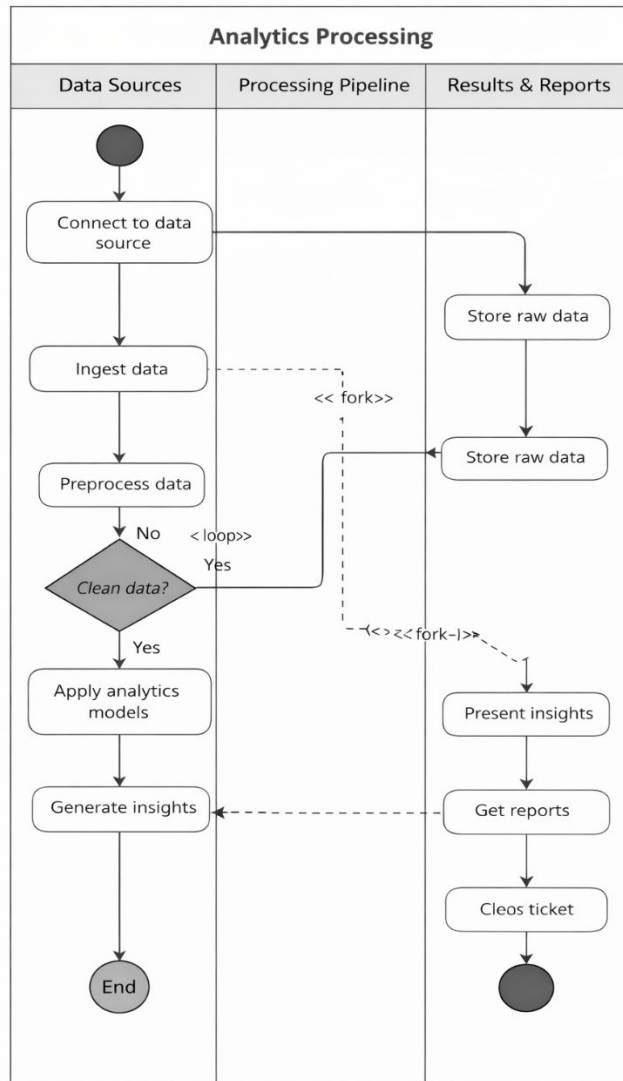
i. Issue Reporting & AI Processing



ii. Ticket Handling Process

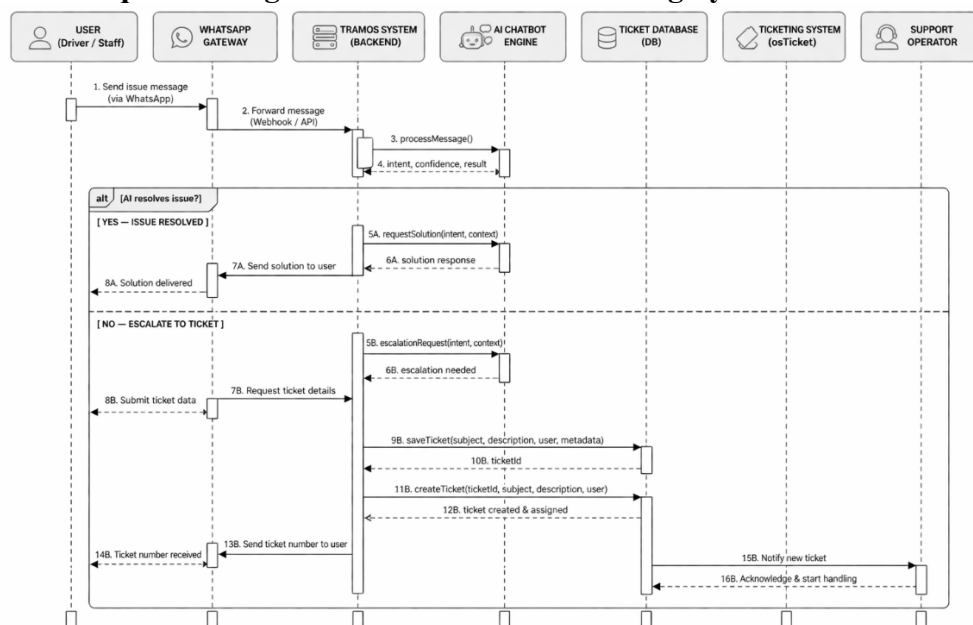


iii. Analytics Processing



4. Sequence Diagram

i. Sequence Diagram: AI Chatbot & Ticketing System



The sequence begins when a user reports a problem by sending a message using WhatsApp. The message is received by the WhatsApp Gateway and then routed to the TRAMOS system using API technology. The system then processes the message and sends it to the AI Chatbot Engine to analyze the intent of the user and categorize the problem.

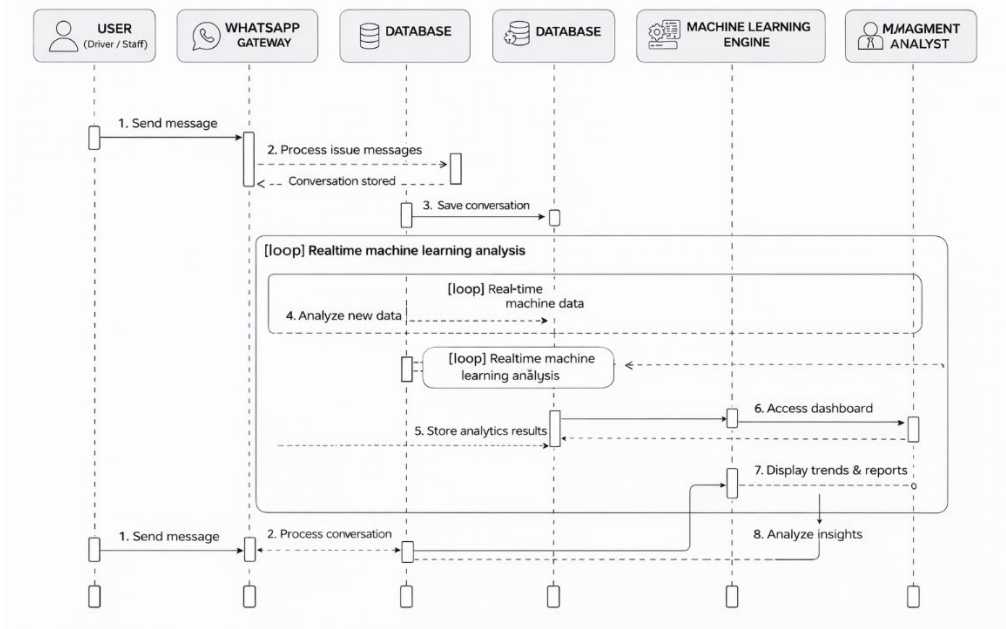
After determining the intent of the user, the system tries to generate a solution automatically by querying the AI system along with context data. If the problem can be solved automatically, then a solution is sent back to the user using WhatsApp.

But in cases where the problem cannot be solved automatically by the system, then a ticketing system is activated by the AI system by asking questions from the user regarding the problem faced by the user. After getting a response from the user regarding the problem, a ticket is created in a structured format and then stored in the database.

Finally, a ticket number is generated and sent back to the user as a confirmation message. At the same time, a message is sent to the support operator regarding the ticket created by the system. The support operator then updates the ticket and solves the problem accordingly.

From the above sequence diagram, we can clearly see how a ticketing system is integrated into the AI system to efficiently resolve problems reported by users in a structured way.

ii. Sequence Diagram: Dashboard & Machine Learning Analytics



The first step in this process is initiated when a message is sent by a user through a WhatsApp message, which is then processed by the system and recorded in the database under a new conversation with the user. Each interaction with the chatbot is continuously recorded, providing a comprehensive record of interactions with users.

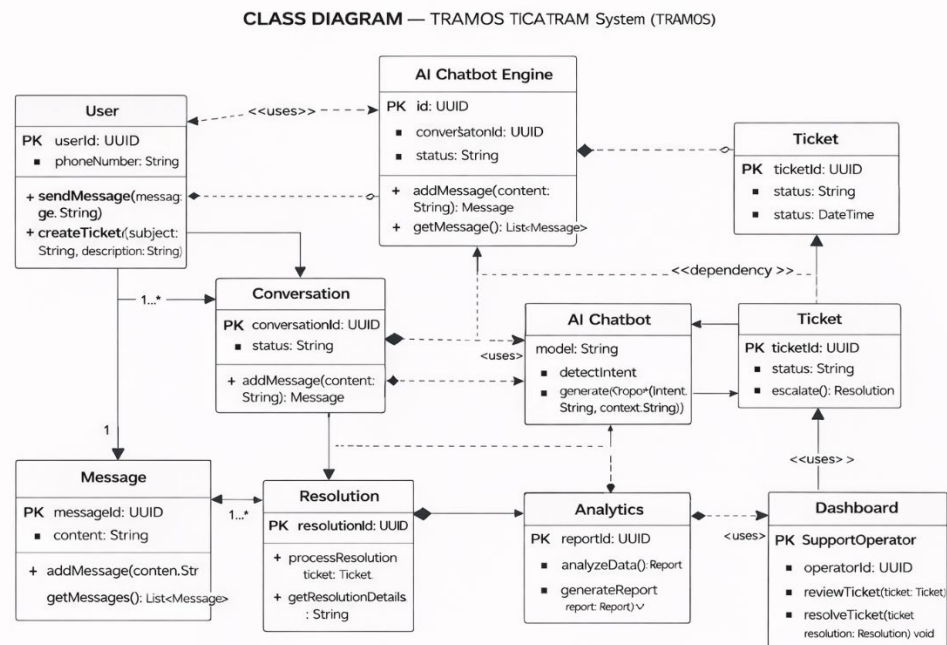
Each time new data is recorded in the system, the machine learning analysis module is automatically initiated to carry out analysis on the data in real-time. The machine learning module processes the data to identify trends in issues, success rates in resolving issues, time taken in resolving issues, and common issues faced by users.

The processed analytical data is then recorded in the system, which can then be viewed on a dashboard system by a management analyst. When a management analyst logs in to the analytics dashboard, the system retrieves the latest data processed by the machine learning module and displays this in a format of charts, trends, and other visual representations.

Using this data, a management analyst can assess the system, identify issues, and make intelligent decisions to enhance system performance, which would otherwise have taken a lot of time and resources.

Thus, through this entire sequence, the system enables a management analyst to carry out real-time analytics and intelligent decision-making through the integration of machine learning and chatbot technologies with a dashboard system.

5. Class Diagram



The class diagram below represents how the major components of the TRAMOS Issue Reporting System work together in the system.

The system has a User class, which represents drivers or staff members who interact with the system by sending messages and creating support tickets. Each User instance can have one or more Conversation instances, which represent the interaction flow with the system.

Each Conversation instance can have multiple Message instances, which represent the interaction of information in a single conversation with a system user.

The messages are processed by the system through the AI Chatbot class, which processes the messages and provides automated responses to the system users. If a problem cannot be solved through automated responses, a Ticket instance is created, which contains information such as status and problem description.

The support ticket is then processed by the Support Operator, who resolves the problem presented in the ticket. The resolution process is represented by the Resolution class, which contains information on the outcome of each ticket handling process.

The system processes all interactions and ticket information through the Analytics class, which generates insights on the interactions in the system. The insights are then presented through the Dashboard, which helps management analysts to analyze system trends.

The class diagram above represents how the system works, with a combination of user interaction, artificial intelligence, ticket management, and analytics, which are integrated in a structured manner in the system.

C. STANDARDS USED

Standards are used as guidelines, requirements, and best practices to ensure quality, security, compatibility, and consistency in the system development process. In this project, various standards are used in terms of system design, system development methodology, technologies used, and data handling.

No	Category	Standard Technology	Description	Implementation in This Project
1	Design Modeling	UML (Unified Modeling Language) 2.x	Standard notation for system modeling diagrams	Used to design Use Case, Activity, Sequence, and Class Diagrams
2	Development Methodology	Agile (Iterative Development)	Flexible development approach with continuous improvement	System developed incrementally with iterative improvements
3	Backend Technology	Python (FastAPI)	High-performance backend framework for API development	Used to build RESTful APIs for chatbot, ticketing, and analytics
4	Frontend Technology	JavaScript / TypeScript	Programming language for frontend/dashboard	Used to build dashboard interface and user interaction
5	AI Technology	AI / NLP Model (OpenAI)	Model for intent detection and response generation	Used to classify issues and generate automated chatbot responses
6	Database	PostgreSQL	Relational database management system	Used to store users, conversations, tickets, and analytics data
7	Communication Platform	Meta WhatsApp Business API	Messaging platform integration via webhook	Used to receive and send user messages in real-time
8	Ticketing System	osTicket	Open-source helpdesk ticketing system	Used to manage support tickets and issue tracking
9	Data Format	JSON	Standard data exchange format	Used for API communication and webhook payloads
10	Character Encoding	UTF-8	Standard text encoding format	Ensures consistent handling of text data
11	Time Standard	ISO 8601	Standard date-time format	Used for timestamps in logs, tickets, and analytics

D. IMPLEMENTATION AND TESTING SCENARIO

The following table lists some simulation scenarios to ensure that the TRAMOS Issue Reporting System works correctly under realistic usage scenarios. The scenarios include actions taken by the user, responses given by the system, and conclusions based on the results.

No	Scenario	Result	Conclusion
1	A user sends an issue report via WhatsApp.	The system receives the message through the WhatsApp Gateway and processes it using the AI engine. The message is stored in the database as part of a conversation.	Confirms successful integration between WhatsApp API, backend system, and database for real-time message handling.
2	The system detects the intent of the user's issue.	The AI model analyzes the message and classifies the issue based on predefined categories (e.g., GPS issue, camera issue).	Validates the AI intent detection process and ensures accurate classification of user issues.
3	The system generates an automated solution.	The AI chatbot retrieves relevant solutions from the knowledge base and sends a response to the user instantly. The response is stored in the system.	Confirms that the AI response generation works correctly and provides real-time assistance to users.
4	The issue is not resolved and escalated to a ticket.	The system offers ticket creation, collects required data, and generates a ticket. The ticket is stored in the database and forwarded to the ticketing system (osTicket).	Validates the escalation mechanism from AI chatbot to ticketing system.
5	Support operator receives and processes the ticket.	The operator receives a notification, reviews ticket details, updates ticket status, and provides resolution. The updates are reflected in the system.	Confirms proper integration between TRAMOS and the ticketing system, ensuring smooth ticket handling.
6	The system sends ticket number and status to the user.	The user receives ticket confirmation and can track the ticket status through the system.	Ensures transparency and real-time communication between system and user.
7	System stores conversation and ticket data for analytics.	All interaction data (messages, tickets, resolutions) are stored in PostgreSQL and prepared for analytics processing.	Validates data persistence and readiness for analytics processing.
8	Management analyst views analytics dashboard.	The system processes stored data using analytics/ML and displays insights such as issue trends, resolution rate, and response time.	Confirms that analytics and dashboard visualization work correctly for decision-making support.

No	Scenario	Result	Conclusion
9	Management analyzes trends and reports.	The analyst interprets system-generated reports and identifies patterns or recurring issues.	Ensures the system supports data-driven decision making and performance evaluation.

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